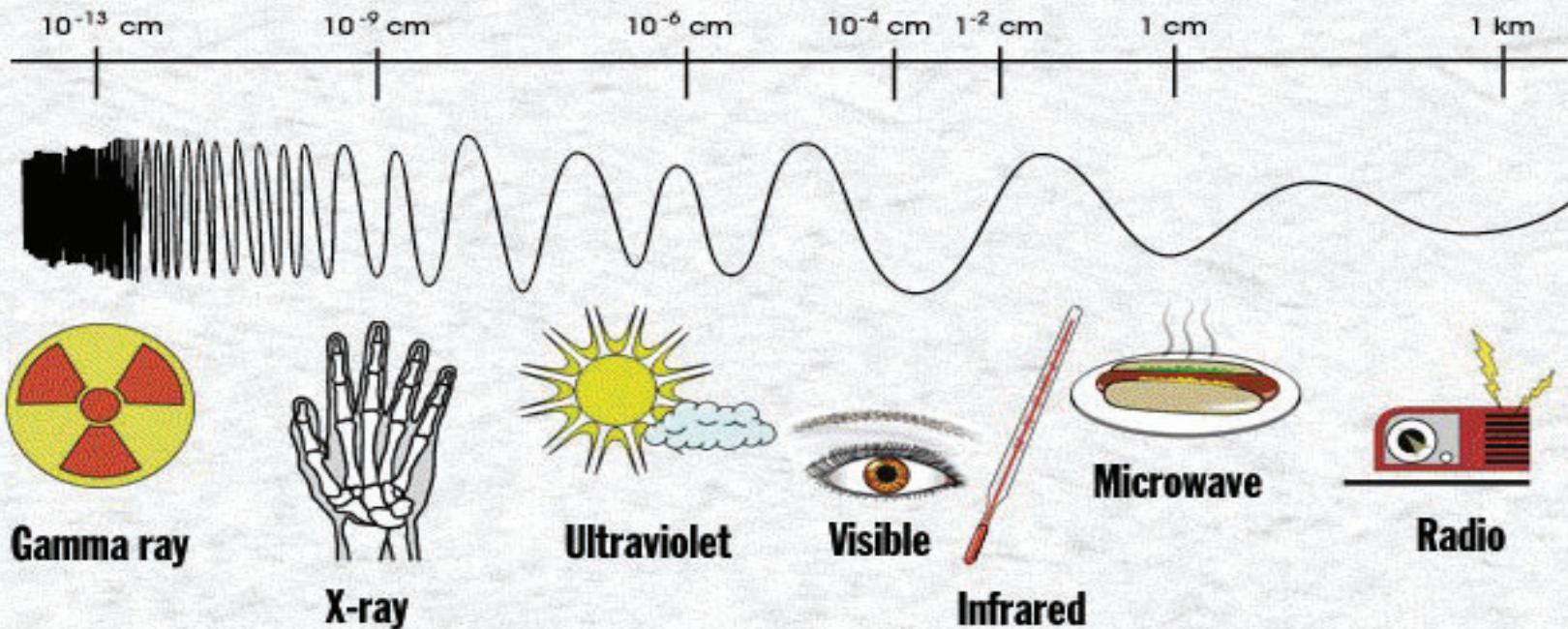
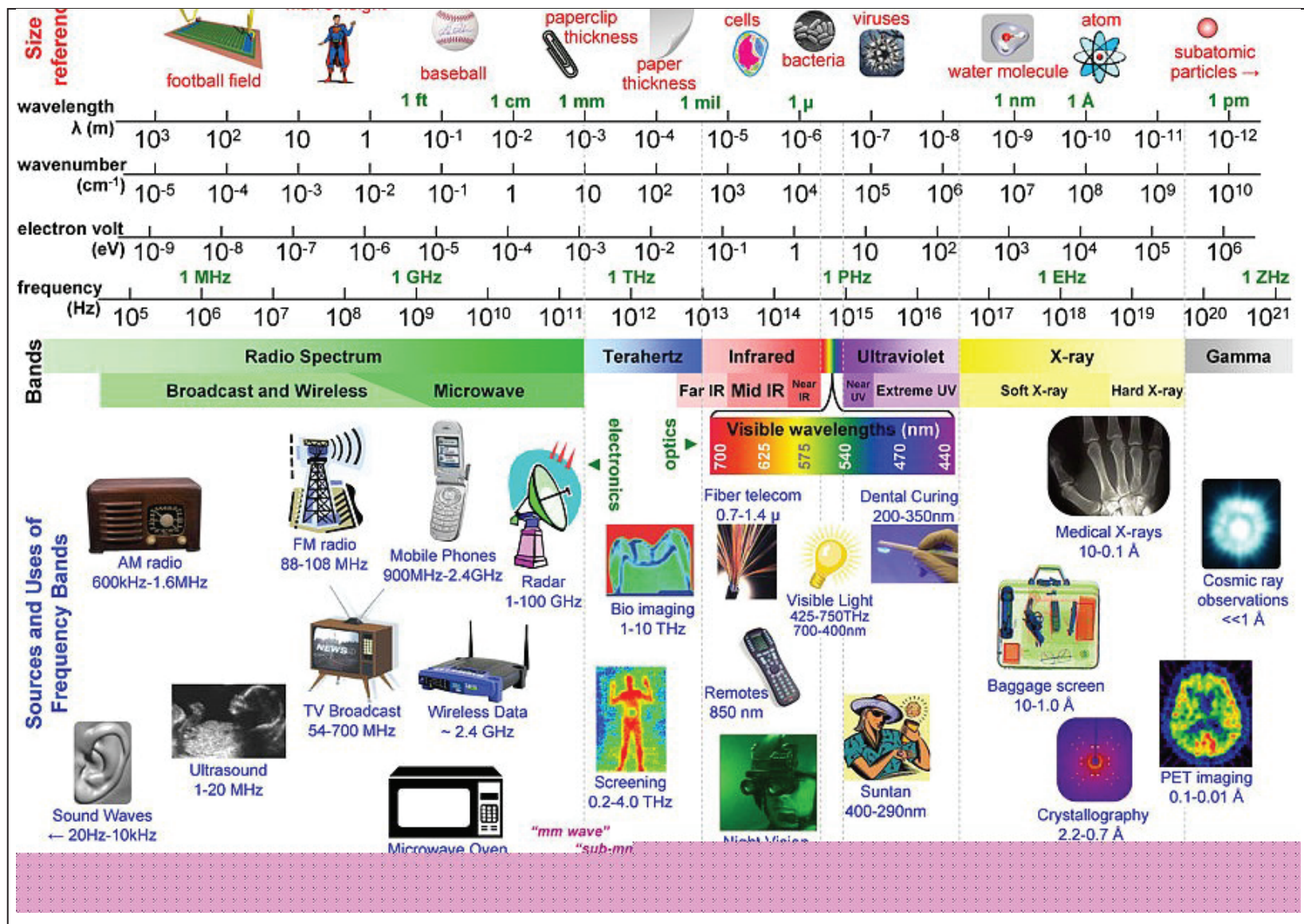


The Electromagnetic Spectrum



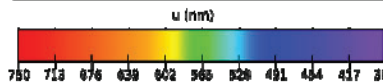


ELECTROMAGNETIC SPECTRUM

(Italy)

FM: 87,5MHz - 108 MHz
AM: 500KHz - 1,5 MHz
Amateur radio:
144MHz - 146MHz

Blue 450-495nm Orange 590-620nm
Green 495-570nm Red 620-750nm



VISIBLE LIGHT
(400nm - 700nm)

Wi-Fi (2,4GHz / 5GHz)
802.11a: ~ 5,2 / 5,8 GHz
802.11b/g: 2412MHz - 2484MHz
802.11n: ~ 2,4 / 5GHz

802.11 b/g Channels	Data rate
1 2412 MHz	Std (Mbit/s)
2 2417 MHz	a 6, 9, 12, 18
3 2422 MHz	a 24, 36, 48, 54
4 2427 MHz	a 1, 2, 5, 11
5 2432 MHz	b 1, 2, 5, 11
6 2437 MHz	b 1, 2, 5, 11
7 2442 MHz	b 1, 2, 5, 11
8 2447 MHz	b 1, 2, 5, 11
9 2452 MHz	b 1, 2, 5, 11
10 2457 MHz	b 1, 2, 5, 11
11 2462 MHz	b 1, 2, 5, 11
12 2467 MHz	b 1, 2, 5, 11
13 2472 MHz	b 1, 2, 5, 11
14 2477 MHz	b 1, 2, 5, 11

ELF 3Hz 30Hz	SLF 30Hz 300Hz	ULF 300Hz 3kHz	VLF 3kHz 30kHz	LF 30kHz 300kHz	MF 300kHz 3MHz	HF 3MHz 30MHz	VHF 30MHz 300MHz	UHF 300MHz 3GHz
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ELF: Extremely Low Frequency
SLF: Super Low Frequency
ULF: Ultra Low Frequency
VLF: Very Low Frequency
LF: Low Frequency
MF: Medium Frequency
HF: High Frequency
VHF: Very High Frequency
UHF: Ultra High Frequency

RADIO FREQUENCIES

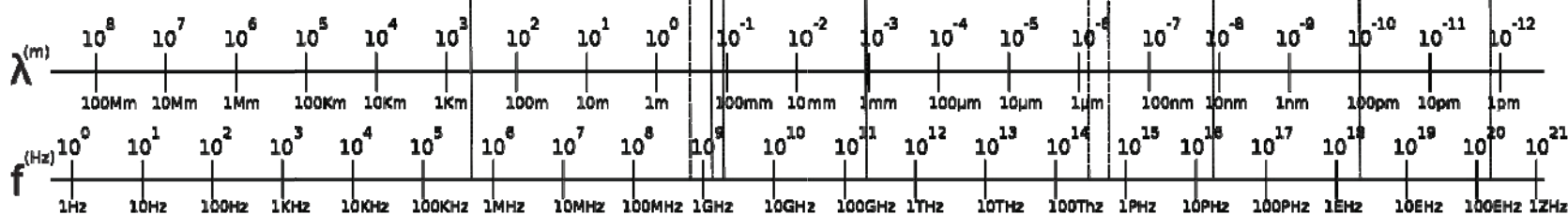
MOBILE PHONES
GSM: 800-1900MHz
Bandwidth: 12.5MHz
Frequency: 890-915 MHz
Frequency: 930-960 MHz
Frequency: 1850-1910 MHz
Frequency: 1930-1990 MHz
UMTS/HSPA: Downlink: 2130-2200MHz Uplink: 1920-1980MHz

Near IR (NIR): 700/750nm - 5µm
Mid IR: 75µm - 25/40µm
Long IR (FIR): 25/40µm - 200/350µm

UV-A: 400-315nm
UV-B: 315-280nm
UV-C: 280-210nm

SOFT X-RAYS λ > 0.1nm
HARD X-RAYS λ < 0.1nm
GAMMA RAYS

POWER & TELEPHONE RADIO & TV MICROWAVES INFRARED UV X-RAYS



SI Prefixes

10 ²⁴	yotta	Y	10 ¹²	tera	T	10 ⁻⁶	micro	µ	10 ⁻¹⁸	atto	a
10 ²¹	zetta	Z	10 ⁹	giga	G	10 ⁻⁹	nano	n	10 ⁻²¹	zepto	z
10 ¹⁸	exa	E	10 ⁶	mega	M	10 ⁻¹²	pico	p	10 ⁻²⁴	yocto	y
10 ¹⁵	peta	P	10 ³	kilo	k	10 ⁻¹⁵	femto	f			
						10 ⁻²	hecto	h			
						10 ⁻¹	deci	d			
						10 ⁻²	centi	c			
						10 ⁻³	milli	m			

Roberto Tacconelli
18-09-2008
/8t4cK